

4. Energy Consumption Report

Title

Feature Scaling in Energy Consumption Data

Objective

The objective of this case study is to apply feature scaling techniques to energy consumption data and evaluate their impact on forecasting models.

Dataset

The dataset used is the Household Electric Power Consumption Dataset obtained from UCI. It contains time-series data of electricity usage including voltage, power consumption, and intensity.

Methodology

The dataset was loaded and cleaned by handling missing values. Relevant numerical features were selected for analysis.

A regression model was first trained without scaling. Then, StandardScaler was applied to normalize the data. The model was retrained on the scaled dataset.

Performance was evaluated using Mean Squared Error (MSE) and results were compared before and after scaling.

Result

The model trained on scaled data showed lower error compared to the unscaled model, indicating improved prediction performance.

Conclusion

This case study shows that feature scaling plays a crucial role in regression models. Proper scaling improves model accuracy and ensures that all features contribute equally to predictions.